1. A regional hospital is preparing for a 5-month surge in cardiovascular procedures. They use highly specialized, single-use Robotic Cardiac Surgical Kits. These kits have a specific sterilization expiration date and can be ordered in quantities of 100, 200, 300, 400, or 500 per month.

   - Each successful surgery using a kit generates a net profit of $100.
   - The Overstock Loss: If a kit is ordered but not used by the end of the month, the specialized components expire, resulting in a $50 loss for the hospital.
   - The Lost Sales Penalty: If a patient arrives for a scheduled robotic surgery and no kit is available, the hospital must perform a traditional surgery or transfer the patient. This logistical scramble and increased recovery time for the patient is valued at a $25 penalty per instance.

   The table provides the distribution of possible demands. Find

   a) the expected value with perfect information (EVwPI);
   b) the best alternative using both EMV and EOL;
   c) the worth of perfect information.

	Outcome				
	1	2	3	4	5
Demand	100	200	300	400	500
Probability	0.2	0.3	0.3	0.1	0.1

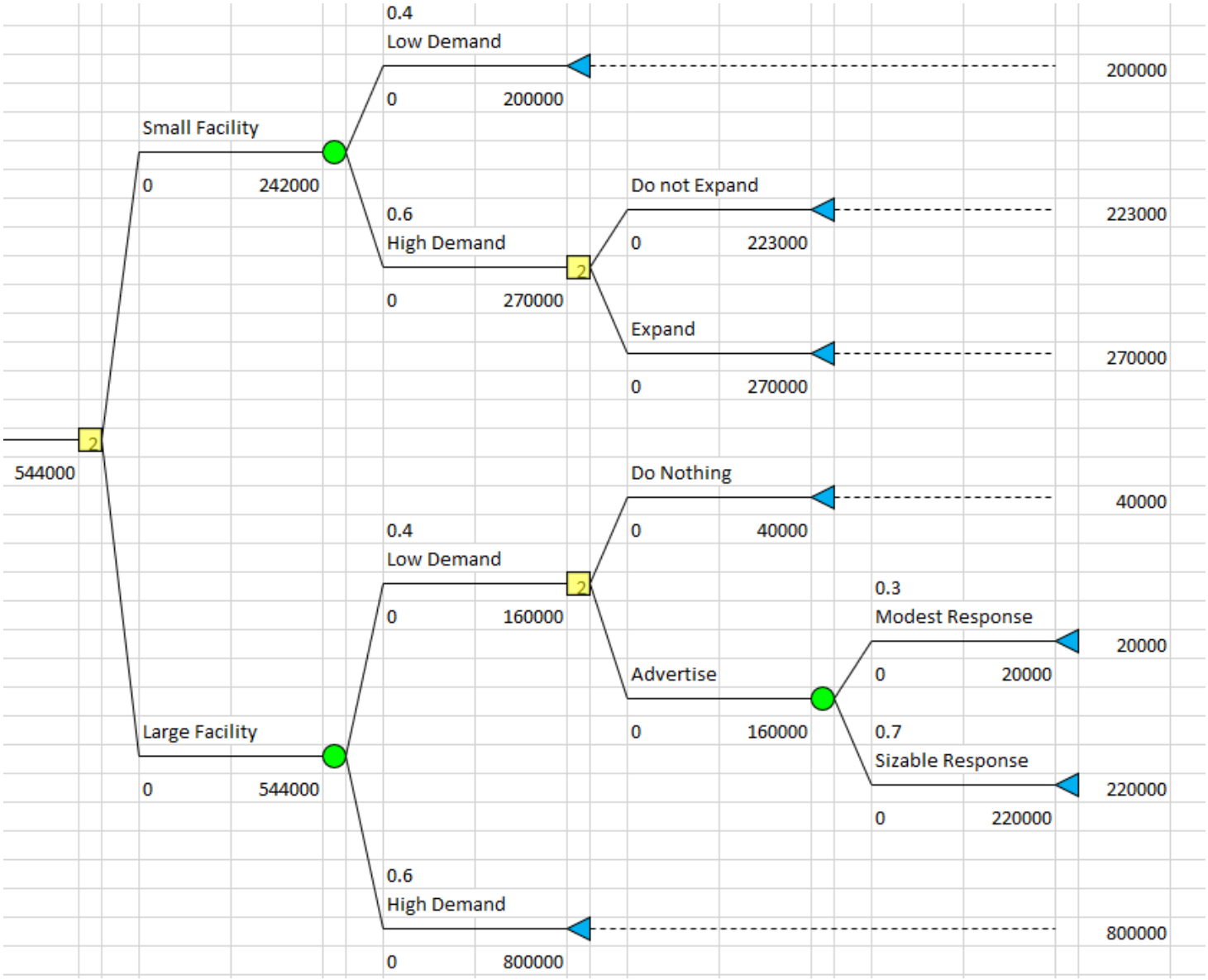
Revenue from sales: $100 \times min(demand, order)$
Cost of unsold: $50 \times max(0, order - demand)$
Cost of shortage: $25 \times max(0, demand - order)$

Payoff Table		Demand (outcomes)					EMV
		100	200	300	400	500	
Alternatives	100	10,000	7,500	5,000	2,500	0	6,000
	200	5,000	20,000	17,500	15,000	12,500	15,000
	300	0	15,000	30,000	27,500	25,000	18,750
	400	-5,000	10,000	25,000	40,000	37,500	17,250
	500	-10,000	5,000	20,000	35,000	50,000	14,000
Probability		**0.2**	**0.3**	**0.3**	**0.1**	**0.1**	

Regrets		Demand (outcomes)					EOL
		100	200	300	400	500	
Alternatives	100	0	12,500	25,000	37,500	50,000	20,000
	200	5,000	0	12,500	25,000	37,500	11,000
	300	10,000	5,000	0	12,500	25,000	7,250
	400	15,000	10,000	5,000	0	12,500	8,750
	500	20,000	15,000	10,000	5,000	0	12,000
Probability		**0.2**	**0.3**	**0.3**	**0.1**	**0.1**	

- $\text{EVwPI} = 0.2(10{,}000) + 0.3(20{,}000) + 0.3(30{,}000) + 0.1(40{,}000) + 0.1(50{,}000) = 26{,}000$
- $\text{EVPI} = \text{EVwPI} - \text{Max EMV} = 26{,}000 - 18{,}750 = 7{,}250$
- $\text{EVPI} = \text{Min EOL} = 7{,}250$
- A perfect forecast of demand is worth no more than $7,250.

2. Either a small or a large facility will be built at a new location. Demand can be either small (prob: 0.4) or large (prob: 0.6). For a small facility and low demand, the payoff is $200,000. Given high demand, we can choose not to expand (payoff $223,000) or to expand (payoff $270,000). For a large facility and high demand, the payoff is $800,000. In low-demand situations, we can choose to do nothing (with a payoff of $40,000) or advertise. Advertising may result in a modest response (prob: 0.3) or a sizable response (prob: 0.7). The payoff will be $20,000 for a modest response and $220,000 for a sizable response. Determine the sequence of decisions.



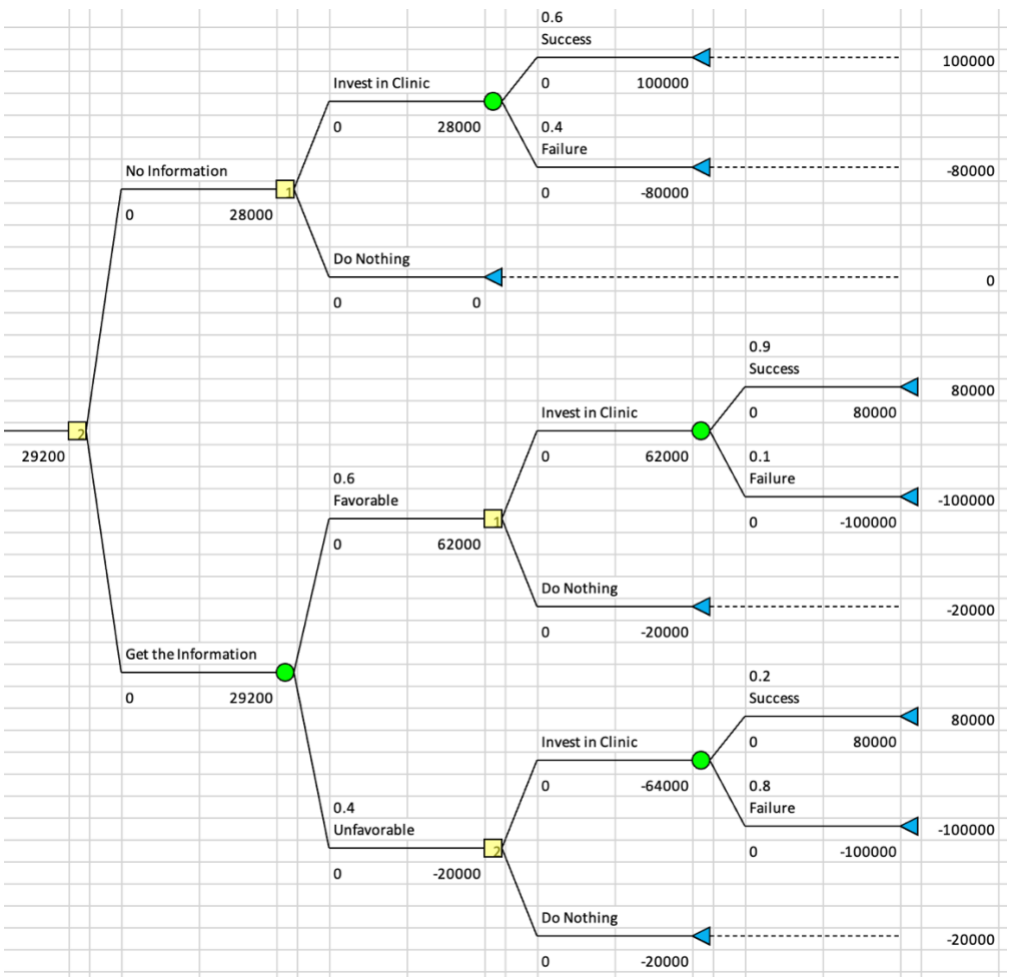
Sequence of decisions: Build a large facility. If low demand happens, advertise.

3. We are deciding whether to invest in a clinic. If we invest in the clinic:
   - If it is successful (prob.=0.6), the payoff is $100,000
   - If it is not successful, a loss of $80,000 will occur.

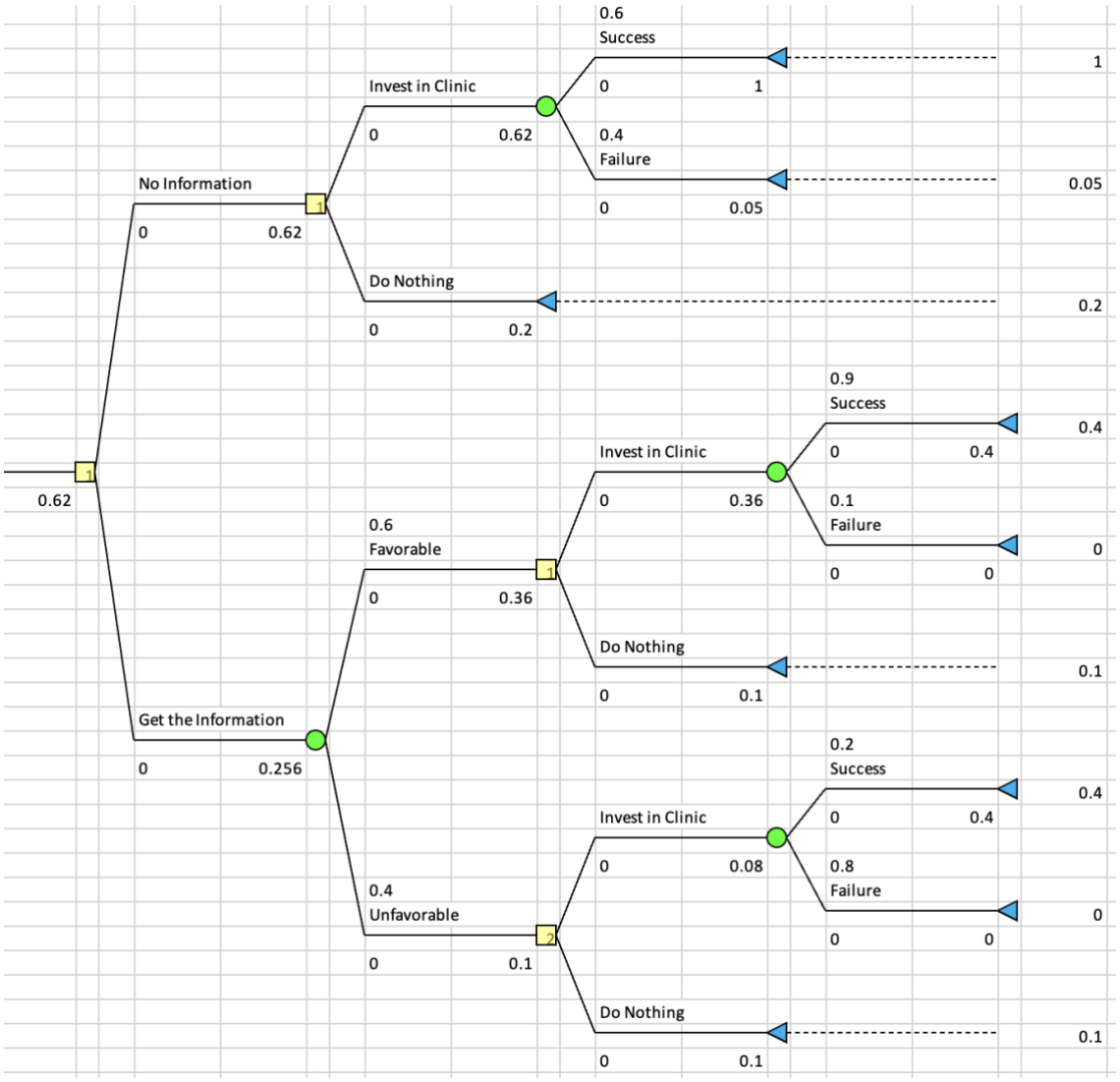
   We try to find more information. A market research for $20,000 is suggested. The information from market research is favorable, with a probability of 60%
   - Favorable information will change the likelihood of success to 0.9
   - Unfavorable information will change the likelihood of success to 0.2

   A) Using payoffs, what is the optimal decision? Calculate the efficiency of the market research.
   B) What is the optimal decision using the utilities below:

Monetary Value	Utility
-$100,000	0
-$80,000	0.05
-$20,000	0.1
$0	0.2
$80,000	0.4
$100,000	1



$$EVSI = 29{,}200 + 20{,}000 - 28{,}000 = 21{,}200$$
$$EVwPI = 100{,}000 \times 0.6 + 0 \times 0.4 = 60{,}000$$
$$EVPI = 60{,}000 - 28{,}000 = 32{,}000$$
$$\text{Efficiency} = \frac{21{,}200}{32{,}000} = 0.6625$$



4. We manage a large outdoor fruit stand in the Halifax Farmer's Market. To replenish the supply, we buy boxes of fruit early each morning from a grower in the Valley. About 85 percent of the fruit is satisfactory, but the other 15 percent is unsatisfactory. A satisfactory box contains 90% excellent fruit and will earn us a $600 profit. An unsatisfactory box contains 40% excellent fruit and will result in a $2,000 loss. Before we decide to accept a box, we are given the opportunity to sample one piece of fruit to test whether it is excellent. Based on that sample, we can then reject the box without paying for it.

   We wonder (1) whether we should continue buying from this grower, (2) if so, whether it is worthwhile sampling just one piece of fruit from a box, and (3) if so, whether we should be accepting or rejecting the box based on this sampling.

   The alternatives are:
   A1: Continue to buy from the grower.
   A2: Not to continue to buy from the grower.
   The states of nature are: **S**atisfactory and **U**nsatisfactory

   The payoff table and EMVs:

Alternatives	State of Nature		EMV
	S	U	
A1	600	-2,000	0.85(600) + 0.15(-2,000) = 210
A2	0	0	0.85(0) + 0.15(0) = 0
Prior Probability	0.85	0.15	

   We continue to buy from the grower.

   Sampling:
   We define:
   B1: Predicting **S**atisfactory based on the sample.
   B2: Predicting **U**nsatisfactory based on the sample.
   Now, we have:

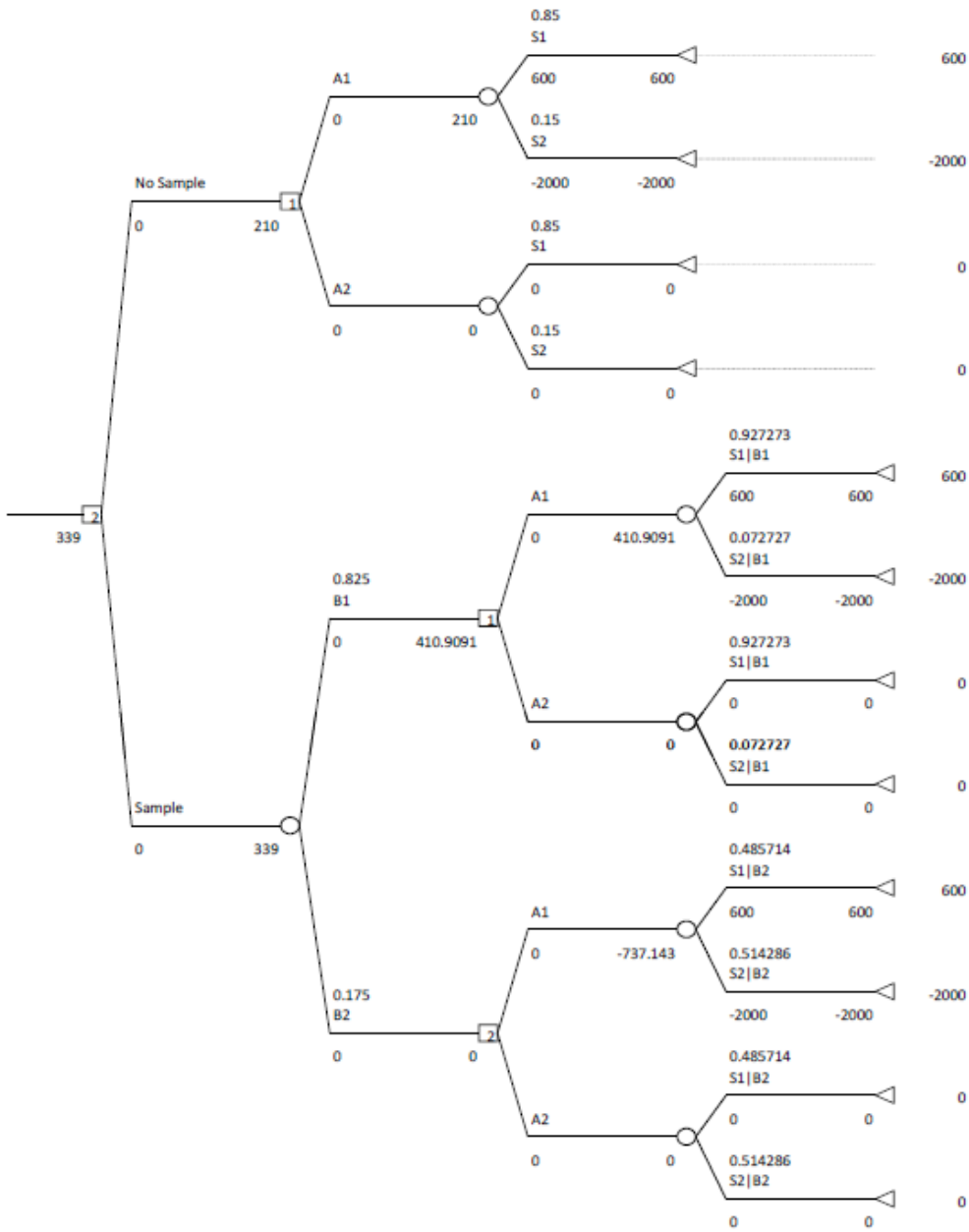
$$P(B1|S) = 0.9 \Longrightarrow P(B2|S) = 0.1$$
$$P(B1|U) = 0.4 \Longrightarrow P(B2|U) = 0.6$$

   Using Bayes theorem, we can calculate the posterior probabilities:

$$P(S|B1) = \frac{P(B1|S).P(S)}{P(B1)} = \frac{P(B1|S).P(S)}{P(B1|S).P(S) + P(B1|U).P(U)} = \frac{0.9 \times 0.85}{0.9 \times 0.85 + 0.4 \times 0.15} = \frac{0.765}{0.825} = 0.927$$

$$P(U|B1) = \frac{P(B1|U).P(U)}{P(B1|U).P(U) + P(B1|S).P(S)} = \frac{0.4 \times 0.15}{0.9 \times 0.85 + 0.4 \times 0.15} = \frac{0.06}{0.825} = 0.0727 = 1 - P(S|B1)$$

$$P(S|B2) = \frac{P(B2|S).P(S)}{P(B2)} = \frac{P(B2|S).P(S)}{P(B2|S).P(S) + P(B2|U).P(U)} = \frac{0.1 \times 0.85}{0.1 \times 0.85 + 0.6 \times 0.15} = \frac{0.085}{0.175} = 0.486$$

$$P(U|B2) = \frac{P(B2|U).P(U)}{P(B2|S).P(S) + P(B2|U).P(U)} = \frac{0.6 \times 0.15}{0.1 \times 0.85 + 0.6 \times 0.15} = \frac{0.09}{0.175} = 0.514 = 1 - P(S|B2)$$

Now, we can have the decision tree. It suggests that sampling is preferable.